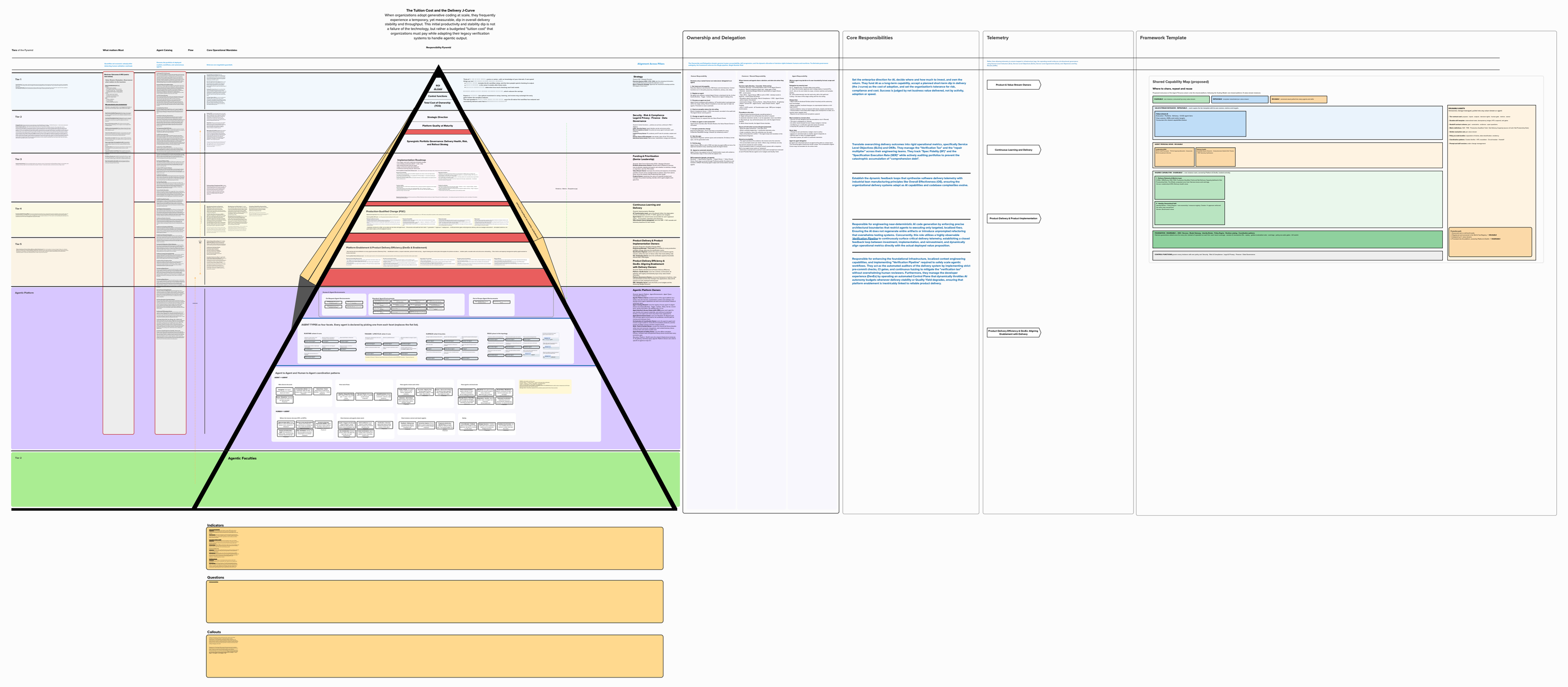
The How:



North Star: Goals, Expectations & Focus

Framework Goals

What the framework is built to achieve. Each goal traces back to a section on this board.

1	**Measure outcomes, not activity**	Tie every metric to an OKR; prove Total Net Productivity Gain (Human + Agent), not usage
2	**Govern every agent**	100% of agents in the Agent Catalog; no Shadow AI
3	**Earn autonomy safely**	High agency inside guardrails; HITL → HOTL as evidence builds
4	**Protect quality, shrink the Verification Tax**	Raise first-pass acceptance; keep rework and AI defects down
5	**Reuse before rebuild**	One shared platform, N value-stream instances; durable skills
6	**Meet in the middle**	Leadership metrics and team adoption converge through middle management
7	**Learn continuously**	Self-improving loop: adopt → measure → learn → codify → scale
8	**Operate at scale**	Reliably and cost-efficiently, within SLOs and token budgets

Telemetry First Platform Solution

Priority #1: set up end-to-end logging before building out the foundational agent platform

You can't govern, improve or prove the ROI of what you can't see. If telemetry isn't in place first, there are no baselines, no Verification Tax, no error budgets, and every later decision is argued from opinion instead of evidence.

1) START HERE
INSTRUMENT: End-to-end logging across prompt → agent → tool → output → review → merge. One trace ID per task.

2) BASELINE
Capture pre-AI workflow time, quality and cost, so Net Time Saved and Verification Tax have a starting point.

3) BUILD THE PLATFORM
Agent Catalog, harness, guardrails and shared capabilities, built on top of live telemetry.

4) SCALE WITH EVIDENCE
Widen autonomy and onboard new waves only when SLOs and error budgets show it's safe.

What "end-to-end" must capture

- **Who / what:** user, team, role, agent ID and version, model, skill invoked
- **Every step:** prompt, context retrieved, tool and MCP calls, outputs, latency, tokens and cost
- **Human touchpoints:** accept / edit / reject, overrides, HITL approvals, review time
- **Outcomes:** build and test result, PR merged, defects traced to AI, rollbacks
- **Safety events:** policy and secret violations, guardrail blocks
- **Correlation:** one trace ID linking all of it, retained per data policy

Exit criteria before the platform build-out

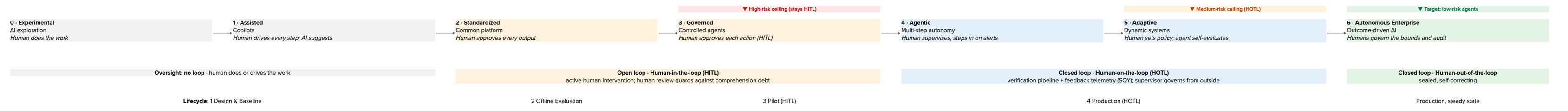
- Nearly all agent runs emit a complete trace (e.g. ⩾ 95%)
- Dashboards live for the Dimensions SLIs, by lens (L / M / IC)
- Baselines captured for every in-scope workflow
- Verification Tax and Net Time Saved computable straight from logs
- PII / secret redaction and retention policy approved
- Alerts wired to SLOs and error budgets

⚠ **Anti-pattern:** launching agents first and "adding observability later." The pre-AI baseline is lost for good, the Delivery J-curve dip looks like failure, and Shadow AI Agents goes unseen.

Maturity Model + Loop Engineering

Purpose: show how each agent moves up the maturity levels as it moves through its development lifecycle. The goal is Level 6 (fully agentic, AI-native), but every agent climbs one level at a time and earns each step with evidence. Risky agents may stop in the middle tiers and keep a human in the loop. Runs on the continuous **Assess → Prioritize → Execute → Measure → Adapt** 90-day transformation loop.

How an agent climbs: one level at a time, each step earned with evidence



▼ High-risk ceiling (stays HITL)
▼ Medium-risk ceiling (HOTL)
▼ Target: low-risk agents

0 · Experimental
AI exploration
Human does the work

1 · Assisted
Copilots
Human drives every step; AI suggests

2 · Standardized
Common platform
Human approves every output

3 · Governed
Controlled agents
Human approves each action (HITL)

4 · Agentic
Multi-step autonomy
Human supervises, steps in on alerts

5 · Adaptive
Dynamic systems
Human sets policy; agent self-evaluates

6 · Autonomous Enterprise
Outcome-driven AI
Humans govern the bounds and audit

Oversight: no loop · human does or drives the work

Open loop · Human-in-the-loop (HITL)
active human intervention; human review guards against comprehension debt

Closed loop · Human-on-the-loop (HOTL)
verification pipeline + feedback telemetry (SQY); supervisor governs from outside

Closed loop · Human-out-of-the-loop
sealed, self-correcting

Lifecycle: 1 Design & Baseline · 2 Offline Evaluation · 3 Pilot (HITL) · 4 Production (HOTL) · Production, steady state

*At every level: **5 Periodic Review** (AI Yield per agent, SLOs, error budget) → **6 Decide**: promote one level · keep · demote · new version vN+1 (re-enters at Design & Baseline) · retire. Stages match **Agent Lifecycle Evaluation**; loop modes match **HITL vs HOTL: Open Loop vs Closed Loop**.*

▲ Promotion gates: move up one level only when
- Offline eval meets the baseline on the agent's eval set (Offline Human Alignment & Analysis)
- SLOs held for a full review period with the error budget intact
- The verification pipeline's staged checks catch failures before output; human review stops finding new failure modes
- AI Yield > 1.0 at Periodic Review
- Telemetry and audit trail complete for every action

No skipping levels: each one produces the evidence the next one needs.

■ Risk ceiling: where an agent stops
- **High risk** (regulated, financial, safety, irreversible actions): ceiling Level 3 Governed. Stays open loop; a human approves every action (HITL)
- **Medium risk** (customer-facing, reversible): ceiling Level 4–5. Closed loop with a human on the loop (HOTL); the supervisor is alerted on SLI breach
- **Low risk** (internal, easily reversed): may reach Level 6, human out of the loop within bounded governance

A ceiling is a design choice, not a failure: a Level 3 agent with high AI Yield is a success.

▼ Demotion: when an agent moves down
- SLO breach or error budget exhausted → drop one level and restore the human checkpoint
- Comprehension debt rising (people can no longer explain what the agent did) → back to HITL
- Incident or policy violation → back to Level 3; review before re-promotion
- Major change to model, prompts or tools → new version vN+1 re-enters at Design & Baseline

Maturity levels in detail

Level	State	Characteristics	Human role · loop mode	Lifecycle stage	Promote to next level when...
0 — Experimental	AI exploration	Individual tools, no governance	Human does the work · no loop	1 Design & Baseline	Use case, owner and baseline metrics defined
1 — Assisted	Copilots	Human-driven, fragmented telemetry	Human drives every step; AI suggests · no loop	1 Design & Baseline	Telemetry captured end to end on the common platform
2 — Standardized	Common platform	Approved models, tools, standards	Human approves every output · open loop (HITL)	2 Offline Evaluation	Offline eval meets baseline; human alignment labels in place
3 — Governed	Controlled agents	Policy-as-code, HITL, audit	Human approves each action · open loop (HITL). Ceiling for high-risk agents	3 Pilot (HITL)	Pilot SLOs met; human review finds no new failure modes; AI Yield > 1.0
4 — Agentic	Multi-step autonomy	Workflows, DAGs, memory, tools	Human supervises, steps in on alerts · closed loop (HOTL)	4 Production (HOTL)	Verification pipeline and supervisor alerts hold SLOs for a full review period
5 — Adaptive	Dynamic systems	Routing, optimization, self-evaluation	Human sets policy; agent self-evaluates · closed loop (HOTL). Ceiling for medium-risk agents	4 Production (HOTL)	Self-evaluation agrees with human audits; error budget intact
6 — Autonomous Enterprise	Outcome-driven AI	Closed-loop execution with bounded governance	Human out of the loop; humans govern bounds and audit · sealed closed loop	Production, steady state	Top level: stays while AI Yield > 1.0 and SLOs hold; otherwise demote

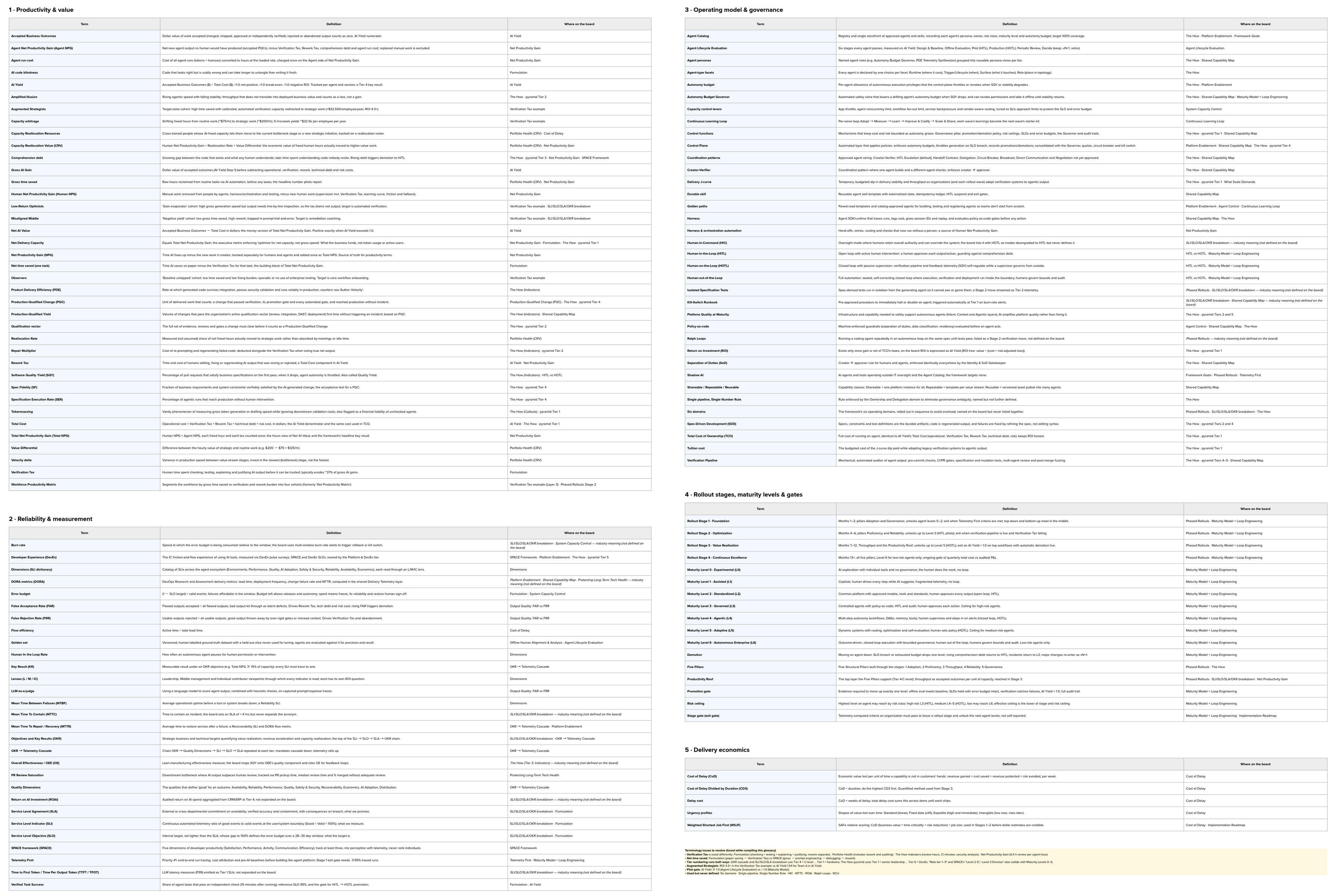
Rollout stages ↔ agent levels: one path to production

The four **rollout stages** (The What → Phased Rollouts) are organization-wide and time-boxed. The seven **maturity levels** apply to each agent. Each stage proves the platform can safely support higher levels and unlocks them. **An agent's ceiling is the lower of its stage ceiling and its risk ceiling. An agent is production-ready when it has passed its own level's promotion gate and the organization has passed the stage gate that unlocks that level.**

Rollout stage (organization)	Timing	Pillars built	Agent levels unlocked	Oversight allowed	Production-readiness exit gate (must pass to enter next stage)
Stage 1 · Foundation	Months 1–3	1 Adoption · 5 Governance	Levels 0–2 (Experimental → Standardized)	No loop, then open loop: human approves every output	Telemetry First exit criteria met: ⩾95% of agent runs traced end to end · pre-AI baselines captured · SLI dashboards live by lens (L / M / IC) · redaction and retention approved · 100% of agents in the Agent Catalog
Stage 2 · Optimization	Months 4–6	2 Proficiency · 4 Reliability	Up to Level 3 (Governed)	Open loop (HITL) pilots: human approves each action	Verification pipeline live (staged checks, specification tests, multi-agent review) · Verification Tax measured and trending down · offline eval suite per agent · HITL pilots hold their SLOs
Stage 3 · Value Realization	Months 7–12	3 Throughput · Productivity Roof	Up to Level 5 (Agentic → Adaptive)	Closed loop (HOTL) for workflows with ⩾95% verified success	AI Yield > 1.0 on the top workflows per function · telemetry joined to CRM / ERP / ITSM · error budgets and Autonomy Budget Governor enforcing demotion automatically
Stage 4 · Continuous Excellence	Months 13+	All five pillars under the Productivity Roof	Level 6 (Autonomous Enterprise), low-risk agents only	Human out of the loop within bounded governance; humans audit	Ongoing, no exit: quarterly total cost vs audited P&L · agents with Verification Tax above time saved retired or re-versioned (vN+1) · full audit trail for every Level 6 agent

Terminology: Terms Used Across the Framework

Every term, acronym and named concept used on this board, defined once and linked to the section that owns it. **Net Productivity Gain is the source of truth for productivity terms.** Entries in *italics* in the last column are used on the board but not defined there; they show the standard industry meaning.



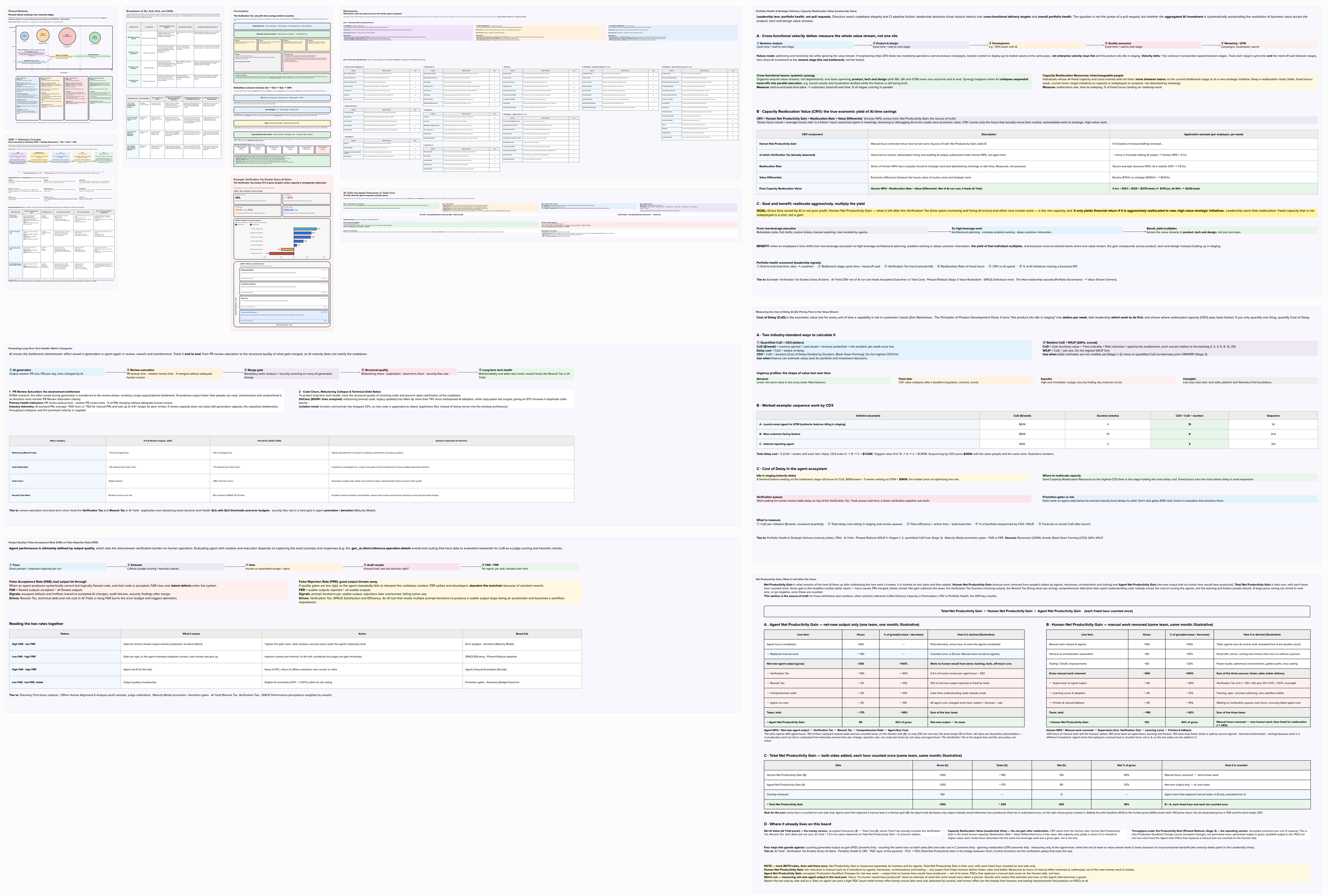
From Pilot to Platform: Scaling Adoption Through Continuous Learning

Continuous Learning Loop

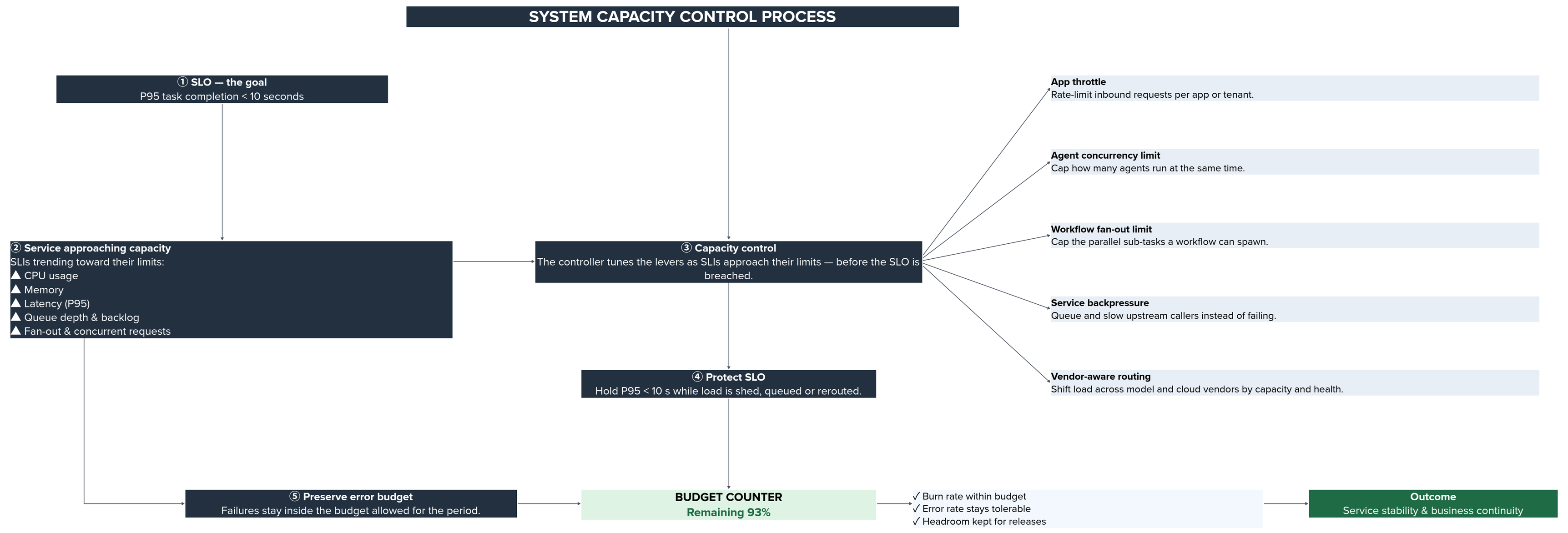
What Scale Demands

ADOPTION
people use it, and keep using it
ROLLOUT
expand to more teams safely
INTEGRATION
fits real workflows and systems
CONTINUOUS LEARNING
the loop gets faster each wave
CHALLENGES
what breaks at scale
RESOURCES
people · budget · time · content
TOOLS
platform and tooling
SUPPORT
operating model and governance

The What: Measuring the frontier agent ecosystem



System Capacity Control: Protecting SLOs & Error Budgets



SYSTEM CAPACITY CONTROL PROCESS

① **SLO — the goal**
P95 task completion < 10 seconds

② **Service approaching capacity**
SLIs trending toward their limits:
▲ CPU usage
▲ Memory
▲ Latency (P95)
▲ Queue depth & backlog
▲ Fan-out & concurrent requests

③ **Capacity control**
The controller tunes the levers as SLIs approach their limits — before the SLO is breached.

App throttle
Rate-limit inbound requests per app or tenant.

Agent concurrency limit
Cap how many agents run at the same time.

Workflow fan-out limit
Cap the parallel sub-tasks a workflow can spawn.

Service backpressure
Queue and slow upstream callers instead of failing.

Vendor-aware routing
Shift load across model and cloud vendors by capacity and health.

④ **Protect SLO**
Hold P95 < 10 s while load is shed, queued or rerouted.

⑤ **Preserve error budget**
Failures stay inside the budget allowed for the period.

BUDGET COUNTER
Remaining 93%

✓ Burn rate within budget
✓ Error rate stays tolerable
✓ Headroom kept for releases

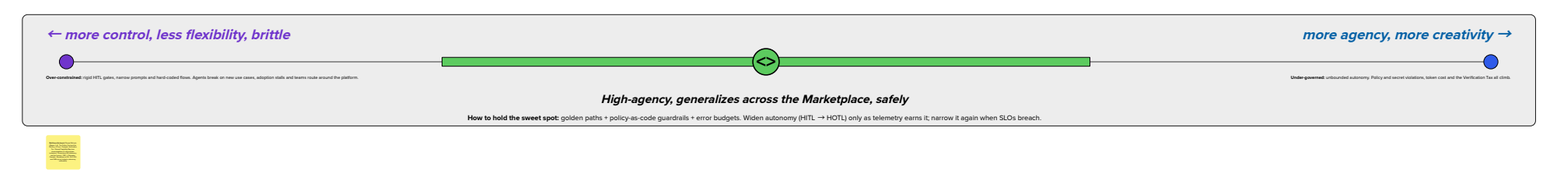
Outcome
Service stability & business continuity

Telemetry measures the SLI. The SLI determines whether the SLO is being achieved. The SLA defines the commitment. Controls protect the SLO/SLA. Error budgets determine how much failure is acceptable before corrective action is required.

Ties to: Breakdown of SLI, SLO, SLA and OKRs (The What) · error budgets + Autonomy Budget Governor (Stage 3 exit gate) · Telemetry First Platform Solution (North Star)

Agent Building

Agent Control: Deterministic vs Creative



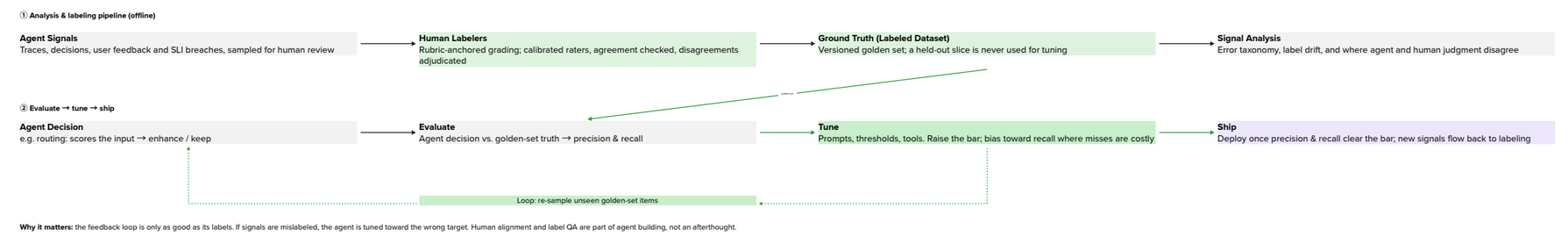
← more control, less flexibility, brittle
more agency, more creativity →

High-agency, generalizes across the Marketplace, safely

How to hold the sweet spot: golden paths + policy-as-code guardrails + error budgets. Widen autonomy (HITL → HOTL) only as telemetry earns it; narrow it again when SLOs breach.

Offline Human Alignment & Analysis

Humans label the signals, so the feedback loop learns the right thing



① Analysis & labeling pipeline (offline)

Agent Signals
Traces, decisions, user feedback and SLI breaches, sampled for human review

Human Labelers
Rubric-anchored grading; calibrated raters, agreement checked, disagreements adjudicated

Ground Truth (Labeled Dataset)
Versioned golden set; a held-out slice is never used for tuning

Signal Analysis
Error taxonomy, label drift, and where agent and human judgment disagree

② Evaluate → tune → ship

Agent Decision
e.g. routing: scores the input → enhance / keep

Evaluate
Agent decision vs. golden-set truth → precision & recall

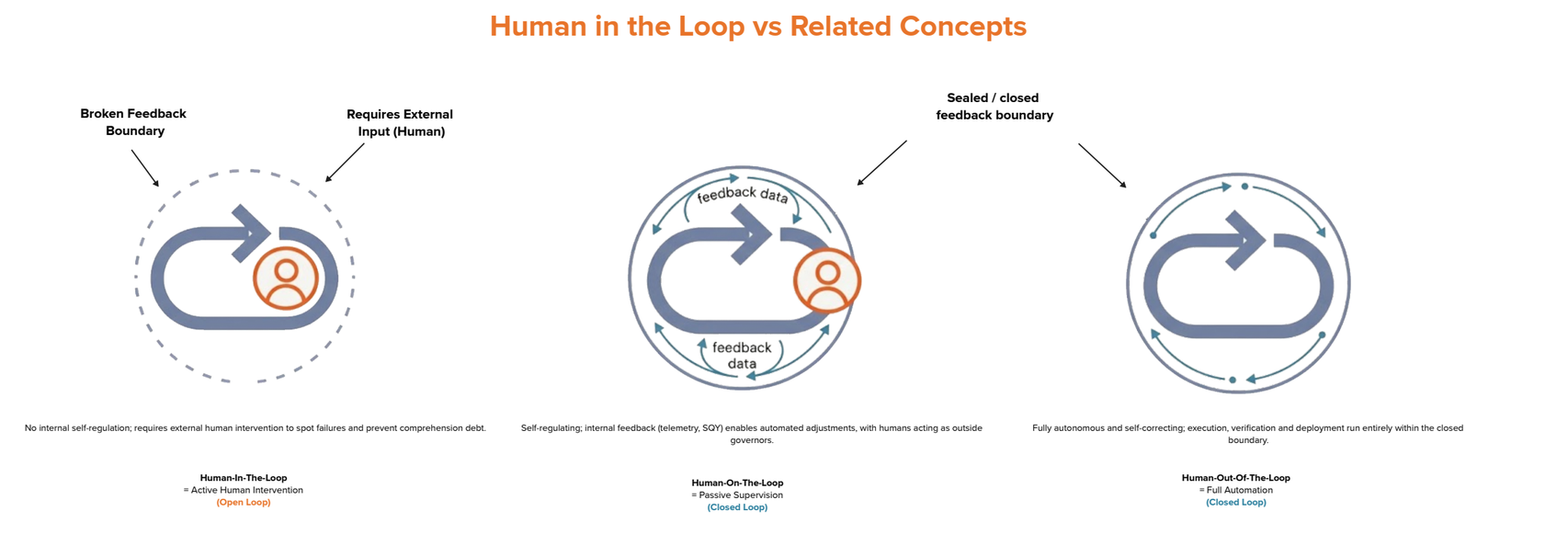
Tune
Prompts, thresholds, tools. Raise the bar; bias toward recall where misses are costly

Ship
Deploy once precision & recall clear the bar; new signals flow back to labeling

Loop: re-sample unseen golden-set items

Why it matters: the feedback loop is only as good as its labels. If signals are mislabeled, the agent is tuned toward the wrong target. Human alignment and label QA are part of agent building, not an afterthought.

HITL vs HOTL: Open Loop vs Closed Loop



Human in the Loop vs Related Concepts
Broken Feedback Boundary
Requires External Input (Human)
Sealed / closed feedback boundary
feedback data
feedback data

No internal self-regulation; requires external human intervention to spot failures and prevent comprehension debt.

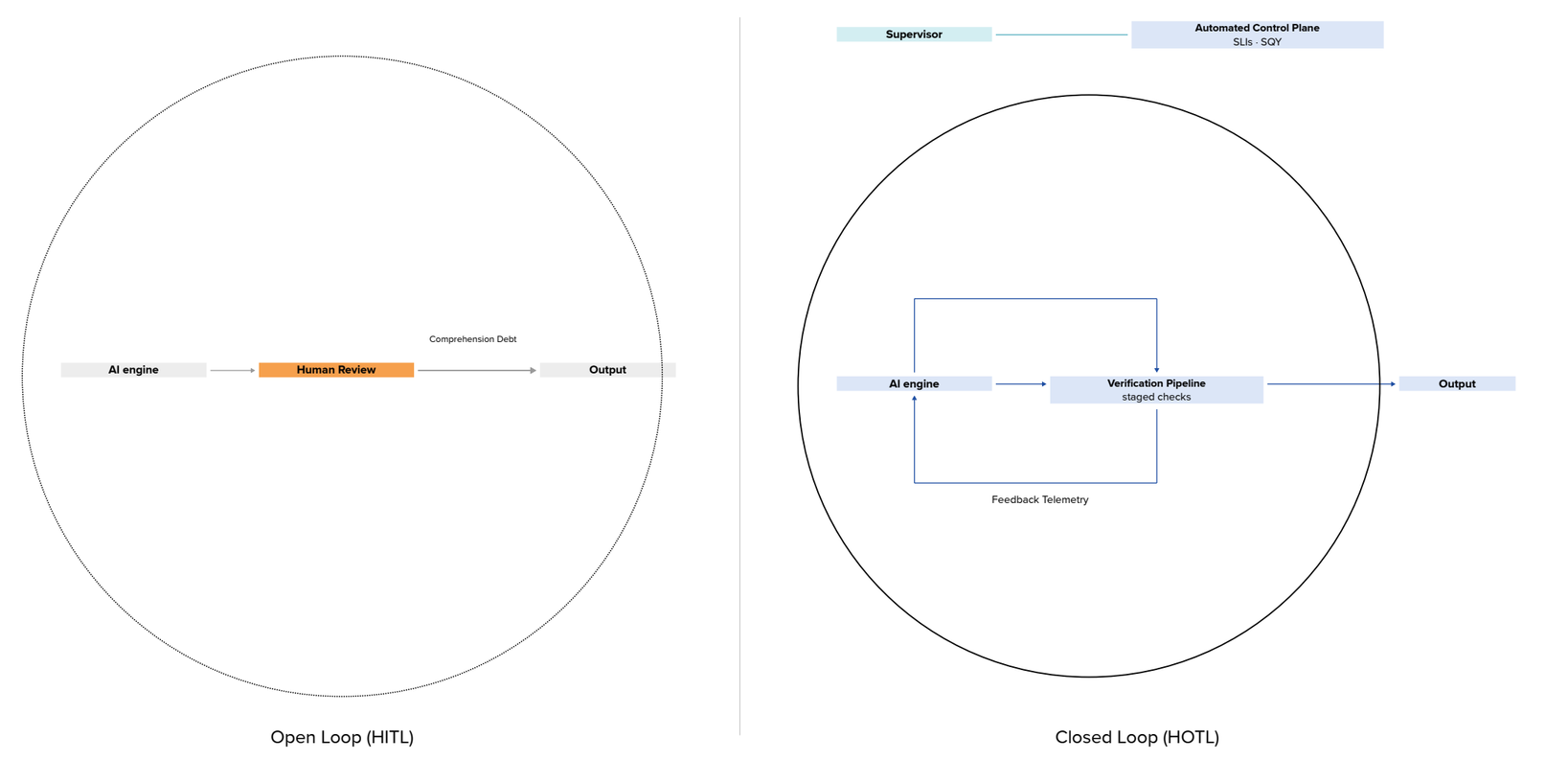
Human-In-The-Loop
= Active Human Intervention
(Open Loop)

Self-regulating; internal feedback (telemetry, SQY) enables automated adjustments, with humans acting as outside governors.

Human-On-The-Loop
= Passive Supervision
(Closed Loop)

Fully autonomous and self-correcting; execution, verification and deployment run entirely within the closed boundary.

Human-Out-Of-The-Loop
= Full Automation
(Closed Loop)



HITL → HOTL: what the transition changes

The HITL trap (Open Loop)
A human is a mandatory step in every cycle: read, edit and approve every PR line by line.
Human attention cannot scale with agent output → the Agentic SDLC Throughput Paradox.
Pays the Verification Tax (review hours, CI overhead, queue time) and accrues Comprehension Debt: reviewers approve code they don't understand to keep the pipeline moving.

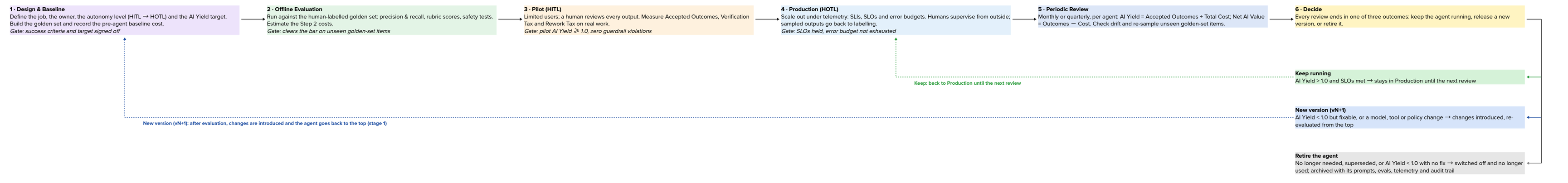
The HOTL target (Closed Loop · agent Levels 4–5)
The agent runs end to end inside predefined boundaries.
The human sits outside the execution path as supervisor, auditor and governor: sets the rules and specs, watches telemetry, intervenes only on a flagged exception or degradation.
Human effort shifts from manual code editing to impact analysis and specification engineering.

How HOTL is enforced
① Spec-Driven Development: humans author specs and constraints; on failure, refine the spec and regenerate rather than hand-edit syntax.
② Mechanical verification pipeline: pre-commit checks, CI gates and mutation-tested suites replace line-by-line review.
③ Automated control plane: if SQY degrades, throttle the autonomy budget or revoke direct merge until a human stabilizes.
④ Specification Execution Rate (SER): % of agent runs reaching production with no human intervention — the proof the loop is closed.

The role shift
From: Syntax Reviewer / Manual Code Editor — a bottleneck inside the loop.
To: Specification Engineer & System Governor — scales the operation from outside the loop.
Human intent still governs autonomous execution, without halting delivery on manual review.
Ties to: Maturity Model (L3 HITL → L4–5 HOTL) · Verification Tax example · SQY · Autonomy Budget Governor

Agent Lifecycle Evaluation

Every agent is evaluated on AI Yield at every stage of its life, not just at launch



1 · Design & Baseline
Define the job, the owner, the autonomy level (HITL → HOTL) and the AI Yield target. Build the golden set and record the pre-agent baseline cost.
Gate: success criteria and target signed off

2 · Offline Evaluation
Run against the human-labelled golden set: precision & recall, rubric scores, safety tests. Estimate the Step 2 costs.
Gate: clears the bar on unseen golden-set items

3 · Pilot (HITL)
Limited users; a human reviews every output. Measure Accepted Outcomes, Verification Tax and Rework Tax on real work.
Gate: pilot AI Yield ⩾ 1.0, zero guardrail violations

4 · Production (HOTL)
Scale out under telemetry: SLIs, SLOs and error budgets. Humans supervise from outside; sampled outputs go back to labelling.
Gate: SLOs held, error budget not exhausted

5 · Periodic Review
Monthly or quarterly, per agent: AI Yield = Accepted Outcomes ÷ Total Cost; Net AI Value = Outcomes − Cost. Check drift and re-sample unseen golden-set items.

6 · Decide
Every review ends in one of three outcomes: keep the agent running, release a new version, or retire it.

Keep running
AI Yield > 1.0 and SLOs met → stays in Production until the next review

Keep: back to Production until the next review

New version (vN+1)
AI Yield < 1.0 but fixable, or a model, tool or policy change → changes introduced, re-evaluated from the top

New version (vN+1): after evaluation, changes are introduced and the agent goes back to the top (stage 1)

Retire the agent
No longer needed, superseded, or AI Yield < 1.0 with no fix → switched off and no longer used; archived with its prompts, evals, telemetry and audit trail

Measured at every stage (from AI Yield): Step 1 Accepted Business Outcomes ($) · Step 2 Total Cost (operational, Verification Tax, Rework Tax, technical debt, risk) · Step 3 AI Yield = Step 1 ÷ Step 2 · Step 4 Net AI Value = Step 1 − Step 2

Rules: each new version starts again at stage 1 and earns its autonomy back · AI Yield is tracked per version, so versions can be compared · retired agents are archived, not deleted

SPACE Framework: Measuring the Individual Contributor & Developer Experience

Level 2 of the operating model. Agent telemetry must be read in the human–computer interaction layer: for any individual contributor (engineer, marketer, analyst) the goal is task execution, flow state and less cognitive load. Measuring raw generation volume (lines of code, words written) rewards bloated repositories and ignores the actual developer experience.

Agent level
Telemetry First: traces, SLIs, cost and verified success per agent

→ **Level 2 · Individual contributor & DevEx ← this section**
SPACE: satisfaction, performance, activity, communication, efficiency

→ **Level 3 · Director: governance, quality & Verification Tax**
PR review saturation, tech-health metrics, AI Yield (The What)

SPACE: five dimensions of developer productivity (Forsgren, Storey et al., 2021). A holistic view of DevEx beyond output metrics.

S · Satisfaction & well-being
How fulfilled and unblocked people feel.
Key driver: cognitive offloading, the degree to which AI absorbs repetitive boilerplate.
Metrics: DevEx pulse survey, perceived productivity, tool satisfaction.
Caution: perception is not velocity; always pair with P.

P · Performance
Outcome quality, not output volume.
Crucial IC metric: acceptance and rejection ratios of AI suggestions, heavily weighted against subsequent rework rates.
Also: escaped defects, hotfixes later in the sprint, net time saved.

A · Activity
Counts of actions completed.
Metrics: AI sessions, suggestions shown / accepted, tasks completed with an agent, PRs opened.
Guardrail: never reward lines or words generated; activity alone is a vanity metric.

C · Communication & collaboration
How work moves between people and agents.
Metrics: PR review pickup time, reviewer load, handoff quality, reuse of shared prompts, skills and agents.
Watch: the shift from creator to reviewer / auditor.

E · Efficiency & flow
Doing work with minimal friction and interruption.
Metrics: time-to-first-draft, context switches (leaving the IDE for a browser AI sandbox), terminal friction, onboarding / transition friction, focus time.
Lever: toolkit and SDK built into the local environment, with cross-tool session continuity.

The paradox: perceived acceleration vs measured deceleration

METR randomized controlled trial: experienced open-source developers using early-2025 AI tools were **19% slower** on complex tasks, while believing they were **20% faster**.
Comprehension debt: when an agent drafts quickly, the human must switch from creator to reviewer and auditor. If the prompt is misaligned with the wider system architecture, the IC spends more time debugging hallucinations and refactoring than writing from scratch.
Comprehension study: developers using AI scored **17 points lower** on a later code-comprehension test (about two letter grades). High acceptance without understanding turns into localized bugs and hotfixes later in the sprint.

So individual productivity is a net number

Net time saved = gross time saved − (prompt engineering + debugging + downstream rework)

Time-to-first-draft alone overstates the gain. Score P (acceptance weighted by rework) next to S (perceived speed): a widening gap between the two is the early warning for comprehension debt, and it flows straight into the Verification Tax and Rework Tax at Level 3. Summed across a team, individual net time saved is the Human side of Total Net Productivity Gain (see *Net Productivity Gain* in The What).

How SPACE fits into this framework

SPACE dimension	Example AI-era metrics	Signal source	Pillar · rollout stage	Feeds into (on this board)
S · Satisfaction & well-being	DevEx pulse, cognitive-offload score, perceived vs measured time saved	Survey (perception)	2 Proficiency · Stage 2	Adoption health in Phased Rollouts; paired with P to expose the perception gap (METR)
P · Performance	Acceptance / rejection ratio weighted by rework; escaped defects and hotfixes; net time saved	VCS + agent telemetry	3 Throughput, 4 Reliability · Stages 2–3	AI Yield (Rework Tax), Protecting Long-Term Tech Health table, agent promotion gates
A · Activity	AI sessions, suggestions accepted, tasks completed with an agent (never lines / words)	Agent + IDE telemetry	1 Adoption · Stage 1	Telemetry First baselines; OKR → Telemetry Cascade SLIs
C · Communication & collaboration	PR pickup and review time, reviewer load, reuse of shared prompts, skills and agents	VCS + catalog	3 Throughput, 5 Governance · Stages 2–3	PR Review Saturation, Verification Tax, Shared Capability Map (reuse before rebuild)
E · Efficiency & flow	Time-to-first-draft, context switches, terminal and onboarding friction, focus time	IDE / SDK telemetry	2 Proficiency · Stage 2	DevEx SLIs / SLOs with error budgets; Continuous Learning Loop (Improve & Codify)

Rules of use: track at least three SPACE dimensions, mixing perception (S) with system telemetry (P, A, E) · never use a single metric or Activity on its own · report at team level, not to rank individuals · a DevEx SLO breach (e.g. rising context switches or rework-weighted rejection) triggers the same Learn → Improve & Codify step as an error-budget burn.